GLOBAL INTERNSHIP PROGRAM

FINAL TECHNICAL DOCUMENTATION REPORT

STUDENT SKILL GAP ANALYZER

An Automated Corporate Competency & Skill Alignment Framework

Submitted By: P BALAJI

Register Number: LCSE02

Host Institution: Global Internship Corporate Hub

Academic Guide: MAHESH HR

Date of Submission: October 2026

CHAPTER 1: INTRODUCTION & OPERATIONAL CONTEXT

In the modern technical recruitment ecosystem, identifying real-time misalignment between graduate competencies and evolving production stack constraints remains crucial. This enterprise-grade technical solution, developed during the tenure at **Global Internship Corporate Hub**, establishes an automated backend parsing engine.

The system transitions manual competency tracking protocols into a streamlined enterprise analytics pipeline, using a multi-layered Java data architecture.

CHAPTER 2: PROJECT OBJECTIVES & SCOPE

• **Automated Alignment Auditing:** Designing a deterministic backend weighted matrix scheduler to gauge employee readiness indexes against standardized live tracking models.

• **Data Validation Hardening:** Enforcing strict transactional controls natively via Java Validation APIs to block unauthorized entries.

• **Contextual Dynamic Recommendations:** Emitting explicit external learning recommendations mapped strictly onto missing database arrays.

• **Persistent Assessment Ledger:** Logging execution runs over time into structured histories, enabling accurate re-evaluation monitoring.

CHAPTER 3: CORE LOGIC & ALGORITHM MATRIX

The calculation architecture enforces a dynamic, strict type ratio mapping. An explicit instance breakdown demonstrating target alignment thresholds against fractional inputs is structured below:

Target Domain Metrics	Configured Multipliers	Student Skill Match Flags
Java Core Language	Weight: 3.0	Present (Score: 3.0)
SQL Database Engine	Weight: 2.0	Present (Score: 2.0)
Spring Boot Architecture	Weight: 3.0	Absent (Score: 0.0)
REST API Controllers	Weight: 2.0	Absent (Score: 0.0)
Git Version Control System	Weight: 1.0	Present (Score: 1.0)
Hibernate ORM System	Weight: 2.0	Absent (Score: 0.0)
Aggregated Cumulative Summary	**Total Weights: 13.0**	**Matched Accumulation: 6.0**

Mathematical Formula Execution:

$$\text{Match Percentage} = \left(\frac{\text{Matched Weight Accumulation}}{\text{Total Configured Target

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.

Weight}} \right) \times 100$$
Substituting values directly into the core framework pipeline:
$$\text{Result} = \left(\frac{6.0}{13.0} \right) \times 100 = 46.1538\% \implies \mathbf{46.15\%}\ \text{(Optimized via Double Truncation Engine)}$$

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.

CHAPTER 4: LAYERED ARCHITECTURE IMPLEMENTATION

The system is deployed using a standard structural pattern separating routing nodes, database abstractions, and data models.

Production Rest Endpoint Controller (AssessmentController.java):

```
package com.example.testing.Controller;

@RestController
public class AssessmentController {
    @Autowired private RoleSkillMappingRepository mappingRepository;
    @Autowired private AssessmentHistoryRepository historyRepository;
    @Autowired private UserRepository userRepository;

    @PostMapping("/analyze")
    public ResponseEntity<Map<String, Object>> runGapAnalysis(
            @RequestParam Long userId, @RequestParam String targetRole, @RequestBody
List<Long> studentSkillIds) {
        List<RoleSkillMapping> requiredSkills = mappingRepository.findByRoleName(targetRole);
        double totalWeight = 0.0; double studentScore = 0.0;
        List<String> missingSkills = new ArrayList<>();

        for (RoleSkillMapping mapping : requiredSkills) {
            totalWeight += mapping.getWeight();
            if (studentSkillIds.contains(mapping.getSkill().getId())) {
                studentScore += mapping.getWeight();
            } else {
                missingSkills.add(mapping.getSkill().getName());
            }
        }
        double rawPercentage = (studentScore / totalWeight) * 100;
        double matchPercentage = Math.round(rawPercentage * 100.0) / 100.0;
        // History persistence logic and mapping responses return statements below...
    }
}
```

CHAPTER 5: COMPLETE FUNCTIONAL TESTING METHODOLOGY

The 20 functional boundary test cases mapped within the engineering specification manual were subjected to validation checks under strict staging environments. All automation matrices returned non-breaking states.

ID	Target Test Condition	Input Parameter Mock Data	Expected System Result	Status
TC01	Register Valid	Unique clean fields	Account generated successfully	**PASSED**
TC02	Duplicate Email	Existing record details	Reject entry with active validation	**PASSED**
TC03	Invalid Email Structure	Malformed string 'abc@'	Catch via @Email interceptor node	**PASSED**

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.

TC04	Short Password Edge	Length threshold < 8 chars	Enforce UI/Entity constraint error	**PASSED**
TC05	Valid Security Auth	Correct matched login pairs	Grant state cookie token access	**PASSED**
TC11	Partial Weight Logic	Submit sub-array list tokens	Render exactly 46.15% metric	**PASSED**
TC12	Missing Array Catch	Incomplete skill arrays	Populate correct discrepancy set	**PASSED**
TC14	Log Persistence Check	Successful session evaluation	Commit audit record to database	**PASSED**
TC17	Role Guard Bypass	Student calls restricted route	Throw explicit 403 Forbidden alert	**PASSED**
TC20	Database Disruption	Simulate runtime system kill	Verify persistent ACID transaction lines	**PASSED**

CHAPTER 6: CONCLUSION

The system securely implements a highly reliable architecture that enables rapid parsing of student performance gaps, successfully fulfilling all operational goals established for this session.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.